SNEHA JHA

Python Developer · AI/ML Engineer · Generative AI

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PROFESSIONAL SUMMARY

Python developer and AI/ML engineer with IEEE-published research and hands-on experience building production-grade RAG pipelines, LLM integrations, and FastAPI backends across FastAPI, LangChain, FAISS, ChromaDB, PostgreSQL, and AWS. Immediate joiner.

EXPERIENCE

Frontend & Backend Developer (Internship) | Jul 2026 – Oct 2026

Blazetrail Fintech Solutions · New Delhi, India (Remote)

- Built frontend features for Kaalfi, a fintech mutual fund investment app, using Next.js (App Router), TypeScript, Tailwind CSS, and shadcn/ui.
- Developed mutual fund scanning and filtering pages, fund detail tabs, portfolio dashboard components, and OTP-based authentication flows.
- Contributed to Django REST Framework backend: auth APIs, Redis-backed token management, and database extensions on PostgreSQL.
- Built PDF factsheet scrapers for 45+ mutual fund companies using BeautifulSoup, Playwright, and GraphQL direct API calls.

Python Developer | Sep 2025 – Jun 2026

Survatra · New Delhi, India

- Sole backend developer. Architected and built a multimodal RAG platform using Python/FastAPI, PostgreSQL, and AWS S3, integrating LLM APIs with vector-based semantic retrieval using FAISS and ChromaDB.
- Built document ingestion pipelines for PDFs, CSVs, and scanned files, including a 4-pass extraction workflow that ran stably on a 1.9GB RAM CPU-only EC2 server.
- Implemented PHI/PII masking before LLM calls, replacing sequential multi-pass regex with a single combined alternation regex for faster processing.
- Implemented PHI data governance with a three-level admin approval system, immutable audit trails, and auto-revoking access windows.
- Built multi-project isolation with scoped API keys, per-project chat history, OTP brute-force protection, and HTTP 429 rate limiting.

AI Developer | Mar 2025 – Jun 2025

The Idea Company (Remote)

- Built an NLP-powered chatbot using LangChain, LLM APIs, and structured knowledge bases to guide startup founders through legal, financial, and operational workflows.

Summer Research Intern, Data Science | Mar 2024 – Jun 2024

IIIT Naya Raipur · Raipur, India

- Developed a DenseNet architecture combining three CNN models, and 1D CNN models using Adam and RMSprop, for COPD detection from pulmonary audio (94% training accuracy with Adam).
- Conducted comparative analysis across 40 papers (2019–2024) and published two peer-reviewed IEEE papers: 15th ICCCNT and IEEE TENSYMP 2024.

AI & ML Developer | Aug 2023 – Sep 2023

Hex N Bit

- Developed AI/ML prototypes during an early-career industry rotation.

Web Developer Intern | Sep 2023

Gauravgo Technologies

- Gained exposure to web hosting, deployment pipelines, and full-stack web fundamentals.

KEY PROJECTS

- **Multimodal RAG Platform** (Survatra, production): document ingestion, PHI/PII masking, FAISS/ChromaDB retrieval, LLM query routing on FastAPI, PostgreSQL, AWS.
- **Hotel Booking RAG Q&A**: FAISS + Sentence Transformers semantic search with a Flask API for natural-language querying.
- **Multi-Agent System**: Python agents with a Gradio interface routing inputs by content type and intent.
- **Stock Price Prediction**: hybrid text + ML models (0.99 accuracy) with Streamlit, Flask, and Grafana dashboards.
- **AI Avatar Conversation Pipeline**: Gemini 2.0 Flash, edge-tts, and Wav2Lip for text-to-speech-to-video output.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS

AI / LLM: LangChain, RAG pipelines, FAISS, ChromaDB, LLM APIs (Anthropic, Gemini, OpenAI-compatible), Qwen (Ollama), Prompt Engineering, NLP

ML / DL: TensorFlow, Keras, 1D CNN, DenseNet, Librosa, scikit-learn, Random Forest

Backend & APIs: FastAPI, Django REST Framework, Flask, REST APIs, Pydantic, Redis

Frontend: Next.js (App Router), React (basics), Tailwind CSS, shadcn/ui

Databases & Cloud: PostgreSQL, AWS S3, AWS EC2, Docker (basic)

Tools: Git, GitHub, Postman, Jira, Streamlit, Gradio, Playwright, BeautifulSoup

EDUCATION

B.Tech Honours in Data Science | 2021 – 2025

Chhattisgarh Swami Vivekanand Technical University

PUBLICATIONS

- “Optimized 1D CNNs for Enhanced Early Detection and Accurate Prediction of COPD and Other Pulmonary Diseases”, 15th ICCCNT, IEEE Xplore, 2024.
- “Enhanced Predictive Modeling Techniques for Early Detection of COPD Utilizing 1D Convolutional Neural Networks”, IEEE TENSYMP, 2024.

CERTIFICATIONS

- Build Website With AI · Getting Started with ReactJS Components · Mobile App Development (Android) · High Performance Scientific Computing · Management Information System